

Student Enrollment & Course Management System (SECMS)

A Salesforce Lightning Application

Project Documentation

Detail	Information
Project Title	Student Enrollment & Course Management System (SECMS)
Platform	Salesforce (Lightning Experience, Developer Org)
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Submitted On	28/09/2026
College / Institution	Government Arts & Science college,vanur

1. Introduction

1.1 Project Overview

The Student Enrollment & Course Management System (SECMS) is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. It enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting within a single, unified Salesforce environment.

The project centralizes all student, course, and enrollment-related data to ensure easy data access, structured workflows, and accurate reporting. Enrollment Officers can manage admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress.

Using Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can handle all enrollment operations, from admission to course completion, systematically and monitor key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose

The primary purpose of SECMS is to replace manual and fragmented enrollment processes with a scalable, automated, and secure Salesforce solution that ensures:

- Accurate student data management
- Structured enrollment workflows
- Automated approvals and validations
- Transparent reporting and monitoring

2. Ideation Phase

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	Manage student enrollments efficiently	Enrollment data is scattered	Systems are manual and unintegrated	Frustrated and error-prone
PS-2	Administrator	Track enrollments and revenue	Reporting is delayed	Data is not centralized	Uncertain in decision-making

2.2 Empathy Map Canvas

User Persona: Enrollment Officer

Quadrant	Description
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Quadrant	Description
Think & Feel	Concerned about double bookings and revenue loss
Hear	Complaints from students and management
See	Multiple spreadsheets and emails
Say & Do	Rechecks data, follows up manually
Pain	Duplicate entries, approval bottlenecks
Gain	Automation, dashboards, faster processing

2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 – Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 – Prioritization

The Salesforce-based SECMS was selected because of its high feasibility and high impact.

3. Requirement Analysis

3.1 Customer Journey Map

The customer journey covers the following stages:

1. Awareness & Registration
2. Course Selection
3. Enrollment Request
4. Fee Payment
5. Approval
6. Instructor Assignment
7. Reporting & Completion

3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, and reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR	Category	Description	Examples
NFR-1	Usability	Simple, intuitive interface usable with minimal training; clear tabs, dashboards, and forms.	Quick registration form; one-click approval; role-based dashboards
NFR-2	Security	Confidentiality and integrity of student data through authentication, authorization, and access control.	Only admins modify courses; instructors see assigned students
NFR-3	Reliability	Consistent, accurate processing of enrollments, approvals, and reports.	No duplicate registration; accurate status updates
NFR-4	Performance	Fast response times, especially during peak admission periods.	Quick dashboard refresh; fast student search
NFR-5	Scalability	Supports growth in students, courses, departments, and users.	Add departments without redesign; extend to portal/mobile
NFR-6	Availability	Continuously accessible during working hours with minimal downtime.	24/7 cloud access; automatic backup

3.3 Data Flow

1. User inputs student and enrollment data
2. System validates and stores records
3. Approval process routes the data
4. Reports and dashboards consume the processed data

In summary, the system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.

3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows

Layer	Technology
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

4. Project Design

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

4.2 Proposed Solution

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation
Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers. Student, Course, and Instructor objects feed into the central Enrollment object, which is governed by validation rules, flows, and an approval process, and is surfaced through reports and dashboards under role-based security.

Data Model

Object	Key Relationships	Purpose
Student__c	Parent of Enrollment__c	Stores student personal, contact, and status information
Course__c	Parent of Enrollment__c	Stores course catalog, category, duration, and fee
Instructor__c	Parent of Enrollment__c	Stores instructor details and expertise
Enrollment__c	Lookups to Student, Course, Instructor	Links students to courses; tracks status, fees, and instructor

5. Project Planning & Scheduling

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

5.2 Product Backlog & Sprint Schedule

Sprint	Epic	USN	User Story / Task	Pts	Priority	Member
1	Developer Setup	USN-1	Create a Salesforce developer account to configure and deploy the system	1	High	Member1
1	Data Modeling	USN-2	Create custom objects: Student, Course, Enrollment, Instructor	3	High	Member2
2	Data Modeling	USN-3	Dedicated tabs for each object for easy navigation	3	High	Member3
2	Data Modeling	USN-4	Create relevant fields (student details, course info, enrollment status)	5	High	Member4
2	Data Modeling	USN-5	Centralized Lightning App for all academic operations	3	High	Member4
2	Data Modeling	USN-6	Define relationships between students, courses, and enrollments	5	Medium	Member1
2	Data Modeling	USN-7	Customized page layouts showing only relevant information	5	High	Member2
3	Automation	USN-8	Validate input data (mandatory fields, enrollment limits)	3	High	Member4
3	Automation	USN-9	Automated flows for enrollment processing and notifications	3	High	Member4
3	Automation	USN-10	Approval process for enrollment requests	5	High	Member1
4	Security	USN-11	Role-based security so users access only relevant data	5	High	Member2
5	Reports	USN-12	Reports on enrollments, course registrations, instructor assignments	4	High	Member3
6	Dashboards	USN-13	Visual dashboards showing enrollment statistics and trends	4	Medium	Member4

5.3 Sprint Tracker & Velocity

Sprint	Total Points	Duration	Start Date	Planned End	Points Completed	Release Date
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022	–	–
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022	–	–
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022	–	–

Velocity = Story Points Completed ÷ Sprint Duration = 20 ÷ 6 ≈ 3.33 points per day.

6. Project Development

6.1 Project Flow (Milestones)

1. Salesforce Developer Account Creation
2. Custom Object Creation (Student, Course, Enrollment, Instructor)
3. Fields Creation
4. Create Relationships
5. Validation Rules
6. Approval Process
7. Tabs Creation
8. Lightning App Creation
9. Page Layouts & Record Types
10. Automation with Flows
11. Reports
12. Dashboards
13. Security Setup
14. Testing & Conclusion

Milestone 1: Salesforce Developer Account

Salesforce is a customer success platform used to sell, service, market, analyze, and connect with customers, storing data securely in the cloud.

Activity 1: Create a Developer Account

1. Go to <https://developer.salesforce.com/signup>.
2. Enter First name, Last name, Email, Role (Developer), Company (College Name), Country (India), and Postal Code.
3. Enter a Username in the format username@organization.com (it need not be a real email).
4. Click Sign me up.

Activity 2: Account Activation

1. Open the verification email (may take 5–10 minutes) and click Verify Account.
2. Set a password, answer a security question, and click Change Password.

3. You are redirected to the Salesforce Setup page.

Milestone 2: Creating Custom Objects

Four custom objects are created: Student, Course, Instructor, and Enrollment. Steps for the Student object:

1. Go to Setup → Object Manager → Create → Custom Object.
2. Label: Student; Plural Label: Students; Record Name: Student Name; Data Type: Text.
3. Under Optional Features select Allow Reports, Allow Activities, and Track Field History (optional).
4. Under Search Status select Allow Search, then click Save.

Repeat the same procedure for Course, Instructor, and Enrollment.

Milestone 3: Fields Creation

Student Object

Field Label	Data Type
Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Course Object

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist (Technical / Language / Non-Technical)
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Instructor Object

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor Code	Text
Expertise	Picklist
Phone	Phone
Email	Email

Enrollment Object

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist (Requested / Approved / Rejected)
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4: Creating Relationships

Three lookup relationships are created on Enrollment__c (Setup → Object Manager → Enrollment__c → Fields & Relationships → New → Lookup Relationship):

Lookup	Related To / API Name	Purpose
Student	Student__c	Link each enrollment to a student
Course	Course__c	Link each enrollment to a course
Instructor	Instructor__c	Store the instructor assigned to the enrollment

For each lookup, complete Field-Level Security and Page Layout options (ensure the field is visible and on the layout), then click Save.

Milestone 5: Validation Rules

Activity 1: Student Email Domain Validation

Purpose: ensure students use only allowed institutional email addresses.

- Object: Student__c; Rule Name: Student_Email_Domain_Validation
- Formula: NOT(CONTAINS(Email__c, "@student.edu"))
- Error Message: Email must be a valid student institutional email.
- Error Location: Field → Email__c

Activity 2: Fees Must Be Paid Before Approval

Purpose: prevent an enrollment from being approved until fees are paid.

- Object: Enrollment__c; Rule Name: Fees_Must_Be_Paid_Before_Approval
- Formula: AND(ISPICKVAL(Enrollment_Status__c, "Approved"), Fees_Paid__c = FALSE)
- Error Message: Fees must be paid before approving an enrollment.
- Error Location: Top of Page

Milestone 6: Approval Process

Purpose: route enrollment requests to the Training Manager for approval or rejection.

1. Go to Setup → Approval Processes → select Enrollment__c → Create New Approval Process → Use Standard Setup.
2. Process Name: Approval_Enrollment_Request; Entry Criteria: Enrollment_Status__c = "Requested".
3. Add an approval step with a specific Training Manager as approver.
4. Final Approval Action: Field Update to set Enrollment_Status__c = "Approved".
5. Final Rejection Actions: Field Update to set Enrollment_Status__c = "Rejected" and an Email Alert to the student.
6. Activate the approval process.

Milestone 7: Tabs

1. Go to Setup → Quick Find → Tabs → Custom Object Tabs → New.
2. Select the Student object, choose a tab style, click Next, assign to all profiles, and Save.
3. Repeat for Courses, Instructors, and Enrollment.

Milestone 8: Lightning App

1. Go to Setup → App Manager → New Lightning App.
2. App Name: Student Enrollment & Course Management; add description and branding (optional).
3. Navigation Style: Standard Navigation.
4. Add items: Student, Course, Instructor, Enrollment, Reports, Dashboards, Home, Tasks.
5. Assign to profiles (Admin, Enrollment Officer, Instructor) and click Save & Finish.

Milestone 9: Page Layouts & Record Types

Activity 1: Record Types for Enrollment

Two record types separate the New Enrollment and Re-Enrollment processes: New Enrollment and Re-Enrollment, both enabled for Admin and Enrollment Officer profiles.

Activity 2: Separate Page Layouts

- New Enrollment Layout: Student, Course, Instructor, Enrollment Date, Status, Fees.
- Re-Enrollment Layout: previous enrollment reference, Re-Enrollment Reason, Status, Fees.
- Use Page Layout Assignment to map each record type to its layout for each profile.

Activity 3: Related Lists on Parent Objects

Add the Enrollments related list to the page layouts of Student__c, Course__c, and Instructor__c so all enrollments are visible from each parent record.

Milestone 10: Automation with Flow

Flow	Trigger	Logic
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Flow	Trigger	Logic
SetEnrollmentDate	Enrollment created	Update Records: set Enrollment_Date__c = \$Flow.CurrentDate
Enrollment Approved Action	Enrollment updated; Status = Approved	Get related Student; send email alert; notify admin; set Student Status = Active
AutoAssign_Instructor	Enrollment created/updated	Get Course; Decision on Category: Technical → Instructor A, Language → Instructor B, Non-Technical → Instructor C; update Instructor__c
Create_Followup_Task	Status = Requested	Get Admin user; create Task 'Follow up on enrollment request' with due date TODAY() + 2, WhatId = Enrollment

Milestone 11: Reports

Report	Report Type	Configuration
Students by Status	Students	Group by Status__c; columns: Name, Email, Phone
Enrollments by Course	Enrollments	Group by Course__r.Name; show record count
Pending Enrollment Approvals	Enrollments	Filter Status = Requested; columns: Student, Course, Owner, Created Date
Revenue Report – Fees Paid	Enrollments	Filter Fees_Paid__c = TRUE; group by Course; summarize fee field

Milestone 12: Dashboards

Create the Student Management Dashboard (Dashboards → New Dashboard) with these components:

Component	Chart Type	Source Report
Student Status	Donut Chart	Students by Status
Enrollments per Course	Bar Chart	Enrollments by Course
Revenue	Gauge	Revenue Report – Fees Paid
Pending Approvals	Table	Pending Enrollment Approvals

Milestone 13: Security Setup

Activity 1: Profiles

- Admin – full access.
- Enrollment Officer – Read, Create, Edit on Enrollment__c; Read (and Edit if required) on Student__c / Course__c.

- Instructor – read-only access to assigned Enrollments.

Activity 2: Permission Sets

- Custom Object Access – additional access to project objects if needed.
- Dashboard Editor – Create/Customize Reports & Dashboards permissions.

Activity 3: Sharing & Organization-Wide Defaults

Object	OWD	Note
Student__c	Public Read/Write	–
Course__c	Public Read/Write	–
Enrollment__c	Private	Instructors see only their own records via a flow that sets OwnerId

7. Functional & Performance Testing

During testing, screenshots were captured from the Salesforce user interface to verify the correct functioning of all implemented features. The following areas were tested:

- Student, course, instructor, and enrollment record creation
- Validation rules for mandatory fields, eligibility, and enrollment constraints
- Record type behavior for New Enrollment and Re-Enrollment
- Flow execution for automated approvals, instructor assignment, and follow-up tasks
- Reports and dashboards for accurate data visualization and enrollment tracking

Test Cases

ID	Scenario	Expected Result	Actual	Status
TC-1	Create student with non-@student.edu email	Validation error displayed	As expected	Pass
TC-2	Approve enrollment with Fees Paid unchecked	Error: fees must be paid	As expected	Pass
TC-3	Create new enrollment	Enrollment Date set to today	As expected	Pass
TC-4	Enrollment for Technical course	Instructor A auto-assigned	As expected	Pass
TC-5	Set status to Requested	Follow-up task created (due +2 days)	As expected	Pass
TC-6	Approve enrollment	Student set to Active; email sent	As expected	Pass
TC-7	Open dashboard	All four components render	As expected	Pass

[Insert screenshots of each test here.]

8. Advantages & Disadvantages

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle, from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. Future Scope

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights
- Mobile access support

11. Appendix

Item	Details
Source Code	Salesforce Configuration
Dataset	Salesforce Objects
GitHub / Demo	Salesforce Developer Org